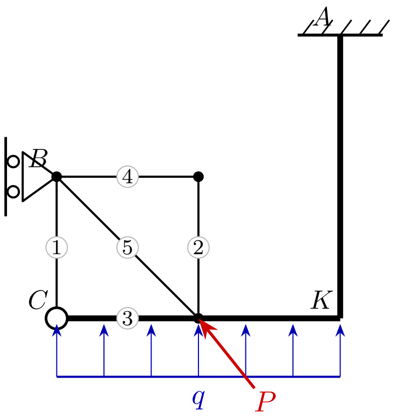
University of Natural Resources and Life Sciences, Vienna (BOKU)

VU Mechanik – LAWI 100515 (PI)

Exam **Group A** Date: 26.06.2026

Model solution



1. Static determinacy

Counting criterion $n = 3k - (a + z)$ (bodies k, reactions a, interaction forces z):

$$n = 3k - (a + z) = 3 \cdot 2 - (4 + 2) = 0 \;\Rightarrow\; \text{statically determinate}$$

2. Support reactions and hinge forces

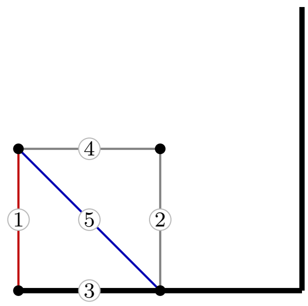
$A: \; R_x = -2, \; R_y = -25, \; M = 46.5 \quad B: \; R_x = 10, \; R_y = 0$

$C: \; H_x = 2, \; H_y = 10$

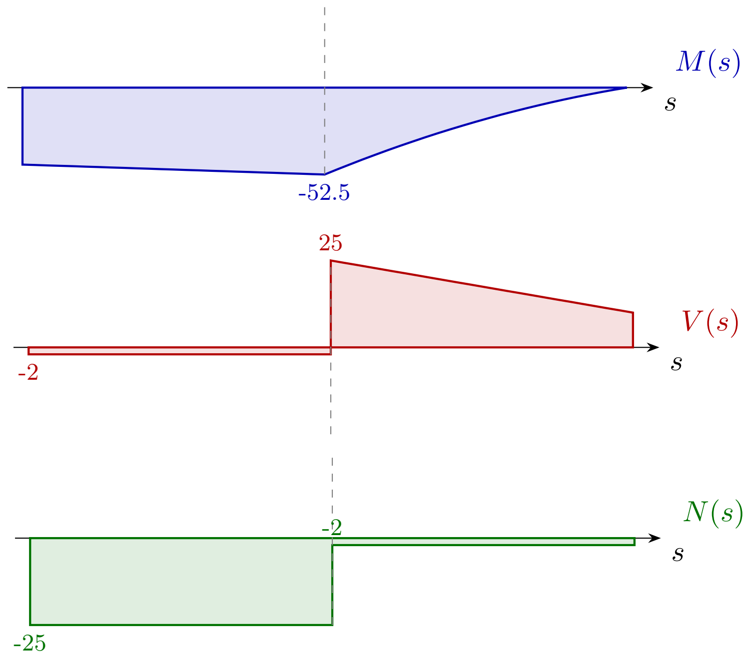
3. Truss member forces

Member	Force S [kN]	Type
1	10	tension
2	0	zero
3	2	tension
4	0	zero
5	-14.14	compression

red: tension (+) blue: compression (−)



4. Section-force diagrams of the beams


M(s)
s
-52.5
25
V(s)
s
-2
N(s)
-2
s
-25